Chapter 12. Nonparametric Regression

Kernel regression, local polynomial regression, bandwidth and order selection, confidence intervals

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Model

$Y_i = m(X_i) + \sigma(X_i)\epsilon_i$, $E\epsilon_i = 0$, $\operatorname{Var}\epsilon_i = 1$, with $m(x) = E(Y \mid X = x)$ smooth but otherwise unrestricted.

12.1 Local polynomial regression

Motivation. Fit a kernel-weighted low-degree polynomial around each point; local linear fits remove the boundary bias of local averages.

Definition (Local polynomial estimator)

With $K_h(u) = K(u/h)/h$, fit at each x

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(Y_i - \beta_0 - \beta_1(X_i - x) - \cdots - \beta_p(X_i - x)^p\big)^2 K_h(x - X_i),$$

and set $\hat{m}_p(x) = \hat{\beta}_{0,x}$. In matrix form, $\hat{\boldsymbol{\beta}}_x = (X_x' W_x X_x)^{-1} X_x' W_x \mathbf{y}$, with $W_x = \operatorname{diag}(K_h(x - X_i))$ and X_x having rows $(1, X_i - x, \ldots, (X_i - x)^p)$.

Each local polynomial estimator is linear: $\hat{m}_p(x) = \sum_i w_i^p(x) Y_i$.

Proposition 12.1

With $S_q = \sum_j K_h(x - X_j)(X_j - x)^q$:

(i) $p = 0$ (**Nadaraya–Watson**): $w_i^0(x) = K_h(x - X_i) / \sum_j K_h(x - X_j)$;

(ii) $p = 1$ (**local linear**): $w_i^1(x) = \dfrac{\{S_2 - (X_i - x)S_1\}K_h(x - X_i)}{S_0 S_2 - S_1^2}$.

In both cases $\sum_i w_i(x) = 1$. For $p = 1$ also $\sum_i w_i^1(x)(X_i - x) = 0$: the fit reproduces straight lines exactly.

Proof [Supplement]. (i) Minimize $\sum (Y_i - \beta_0)^2 K_h(x - X_i)$.

(ii) $X_x' W_x X_x = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, whose inverse has first row $(S_2, -S_1)/(S_0 S_2 - S_1^2)$; multiply by $X_x' W_x$. Then $\sum_i (S_2 - (X_i - x)S_1)K_h = S_2 S_0 - S_1^2$ and $\sum_i (S_2 - (X_i - x)S_1)(X_i - x)K_h = S_2 S_1 - S_1 S_2 = 0$. ■

As $h \to \infty$, Nadaraya–Watson tends to $\bar{Y}$ and local linear to the LS line.

Theorem 12.2

Let m be twice continuously differentiable, let the design density f be smooth with $f(x) > 0$, and use the Gaussian kernel. As $n \to \infty$, $h \to 0$ and $nh \to \infty$, conditionally on the X's and for $p = 0, 1$,

$$\operatorname{bias}\hat{m}_p(x) = B_p(x)h^2 + o_P(h^2), \qquad \operatorname{Var}\hat{m}_p(x) = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \cdot \frac{1}{nh} + o_P\Big(\frac{1}{nh}\Big),$$

with $B_0 = \dfrac{m''}{2} + \dfrac{m' f'}{f}$ and $B_1 = \dfrac{m''}{2}$.

Sketch. Since $\sum_i w_i = 1$, bias $= \sum_i w_i(x)\{m(X_i) - m(x)\}$ and Var $= \sum_i w_i^2 \sigma^2(X_i)$. For $p = 0$ the numerator $\frac{1}{n}\sum K_h(x - X_i)\{m(X_i) - m(x)\}$ has mean $\int K(u)\{m(x - hu) - m(x)\}f(x - hu)\,du = h^2(\frac{m''f}{2} + m'f') + o(h^2)$ and the denominator tends to $f(x)$, giving B_0. Variance: $\frac{1}{n^2}\sum K_h^2\sigma^2 \approx \frac{\sigma^2 f}{nh}R(K)$, $R(\phi) = \frac{1}{2\sqrt{\pi}}$, divided by f^2. For $p = 1$, $\sum w_i^1(X_i - x) = 0$ removes the term $m'(x)(X_i - x)$ exactly, including the design asymmetry behind $m'f'/f$; the variance is unchanged. ■

So local linear has the Nadaraya–Watson variance without the **design bias** $m'f'/f$, and behaves better at the boundary: the default. Bias decreases with degree, $|\text{bias}\,\hat{m}_0| \gtrsim |\text{bias}\,\hat{m}_1| \gtrsim |\text{bias}\,\hat{m}_2| \gtrsim \ldots$, while

$$\text{Var}\,\hat{m}_0 \approx \text{Var}\,\hat{m}_1 < \text{Var}\,\hat{m}_2 \approx \text{Var}\,\hat{m}_3 < \cdots .$$

so odd degrees ($p = 1$, sometimes 3) are preferred. The bias $\frac{m''}{2}h^2$ is positive in valleys and negative at peaks: features are flattened.

12.2 Choosing the bandwidth

Motivation. Cross-validation estimates prediction error, from a single fit for linear smoothers.

Integrating bias and variance against the design density,

$$\text{MISE} \approx \frac{h^4}{4}\int (m'')^2 f + \frac{\int \sigma^2}{2\sqrt{\pi}\,nh}, \qquad h^* = \Big(\frac{\int \sigma^2}{2\sqrt{\pi}\int (m'')^2 f \cdot n}\Big)^{1/5}.$$

- **Direct plug-in** (Ruppert–Sheather–Wand): estimate $\int (m'')^2 f$ and $\int \sigma^2$ by blockwise quartic fits.
- **Leave-one-out cross-validation:** minimize $\text{CV}(h) = \frac{1}{n}\sum_i \{Y_i - \hat{m}^{(-i)}(X_i)\}^2$.

Proposition 12.3

For a linear smoother whose leave-one-out fit equals the full fit at X_i after deleting point i (true for local polynomial fits), $\text{CV}(h) = \dfrac{1}{n}\sum_i \Big(\dfrac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}\Big)^2$.

Proof [Supplement]. Let $\tilde{Y}_i = \hat{m}^{(-i)}(X_i)$. Replacing Y_i by $\tilde{Y}_i$ leaves the fit at X_i unchanged (the added point lies on the leave-one-out fit), so $\tilde{Y}_i = \sum_{j \neq i} w_j(X_i)Y_j + w_i(X_i)\tilde{Y}_i$, i.e. $\tilde{Y}_i = \frac{\hat{m}(X_i) - w_i(X_i)Y_i}{1 - w_i(X_i)}$ and $Y_i - \tilde{Y}_i = \frac{Y_i - \hat{m}(X_i)}{1 - w_i(X_i)}$. ■

So CV needs one fit per h.

12.3 Confidence intervals

Intervals target $m_h(x) = E\hat{m}_1(x)$.

- **Theory-based:** $\widehat{\text{se}}(\hat{m}_1(x)) = \big(\sum_i w_i^1(x)^2\hat{\sigma}^2(X_i)\big)^{1/2}$, $m_h(x) \in \hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}$. Homoscedastic, with smoother matrix W ($\hat{\mathbf{y}} = W\mathbf{y}$):

  $$\hat{\sigma}^2 = \frac{\|(I - W)\mathbf{y}\|^2}{\text{tr}\{(I - W)'(I - W)\}}.$$

  Heteroscedastic: smooth $\log r_i^2$ on X_i and exponentiate.
- **Bootstrap:** resample pairs, refit with the *same* h on a grid; use $\hat{m}_1(x) \pm z_{\alpha/2}\widehat{\text{se}}^*(x)$ or $(Q^*_{\alpha/2}(x), Q^*_{1-\alpha/2}(x))$. Residual bootstraps need care (residuals inherit smoothing bias).

Simultaneous bands (Section 11.4) are rarely used.

12.4 Selected exercises

Motivation. Infinite-bandwidth limits, exact unbiasedness of local linear fits for lines, and when $\hat{\sigma}^2$ is unbiased.

Exercise 12.1, 12.4 (Infinite bandwidth)

Let K be continuous at 0 with $K(0) > 0$ (e.g. Gaussian), and suppose the X_i are not all equal. Show that as $h \to \infty$, for every x:
(a) the Nadaraya–Watson estimate tends to $\bar{Y}$;
(b) the local linear estimate tends to the least squares line $\bar{Y} + \hat{\beta}(x - \bar{X})$, $\hat{\beta} = S_{XY}/S_{XX}$.

Solution.

Step 1. (Scale invariance.) In Proposition 12.1, w_i^0 and w_i^1 are ratios of homogeneous expressions (degrees 1 and 2) in the $K_h(x - X_j)$, so replace these by $\kappa_j(h) = K((x - X_j)/h) \to K(0) > 0$.

(a)

Step 2. $w_i^0 = \kappa_i / \sum_j \kappa_j \to K(0)/(nK(0)) = 1/n$, so $\hat{m}_0(x) \to \bar{Y}$.

(b)

Step 3. With $d_j = X_j - x$, $S_q \to K(0) \sum_j d_j^q$ and the denominator tends to

$$K(0)^2 \Big\{ n \sum_j d_j^2 - \Big(\sum_j d_j \Big)^2 \Big\} = K(0)^2 \, n S_{XX} > 0.$$

So $w_i^1 \to (\sum_j d_j^2 - d_i \sum_j d_j)/(nS_{XX})$.

Step 4. With $\delta = \bar{X} - x$: $\sum_j d_j = n\delta$, $\sum_j d_j^2 = S_{XX} + n\delta^2$, $\sum_i d_i Y_i = S_{XY} + n\delta\bar{Y}$, so

$$\sum_i w_i^1 Y_i \to \frac{(S_{XX} + n\delta^2)\, n\bar{Y} - n\delta(S_{XY} + n\delta\bar{Y})}{nS_{XX}} = \bar{Y} - \delta \frac{S_{XY}}{S_{XX}} = \bar{Y} + \hat{\beta}(x - \bar{X}).$$

■

Exercise 12.5 (No bias for straight lines)

Suppose $m(x) = \alpha + \beta x$. Show that, conditionally on the X's, the local linear estimator is exactly unbiased for every h and x, while the Nadaraya–Watson estimator has bias $\beta \sum_i w_i^0(x)(X_i - x)$, which is nonzero in general (e.g. at $x = \min_i X_i$). Relate this to B_0 and B_1 in Theorem 12.2.

Solution.

Step 1. The weights depend only on the X's, so with $\sum_i w_i^p = 1$

$$E(\hat{m}_p(x) \mid X) - m(x) = \sum_i w_i^p(x) m(X_i) - m(x) = \beta \sum_i w_i^p(x)(X_i - x),$$

Step 2. For $p = 1$, $\sum_i w_i^1(x)(X_i - x) = 0$ by Proposition 12.1, so the bias is exactly 0.

Step 3. For $p = 0$ the bias is $\beta \sum_i K_h(x - X_i)(X_i - x) / \sum_i K_h(x - X_i)$. At $x = \min_i X_i$ all $X_i - x \geq 0$, some > 0, so the bias has the sign of β: the fit is pulled toward the interior.

Consistent with Theorem 12.2: $B_1 = m''/2 = 0$ for a line, while $B_0 = \beta f'/f$ is the design bias. ■

Exercise 12.6 (When is $\hat{\sigma}^2$ unbiased?)

Let $\hat{\mathbf{y}} = W\mathbf{y}$ be a linear smoother with $E(\mathbf{y} \mid X) = \mathbf{m} = (m(X_i))_i$ and $\mathrm{Var}(\mathbf{y} \mid X) = \sigma^2 I$. Show that

$$E(\|(I - W)\mathbf{y}\|^2 \mid X) = \|(I - W)\mathbf{m}\|^2 + \sigma^2 \mathrm{tr}\{(I - W)'(I - W)\}.$$

Deduce that the estimator $\hat{\sigma}^2$ of Section 12.3 is unbiased when m is linear and W is the local linear smoother, and biased upward otherwise.

Solution [Supplement].

Step 1. With $\mathbf{y} = \mathbf{m} + \boldsymbol{\epsilon}$ and $A = (I - W)'(I - W)$,

$$\|(I - W)\mathbf{y}\|^2 = \|(I - W)\mathbf{m}\|^2 + 2\mathbf{m}'A\boldsymbol{\epsilon} + \boldsymbol{\epsilon}'A\boldsymbol{\epsilon}.$$

Step 2. The middle term has mean 0, and $E\boldsymbol{\epsilon}'A\boldsymbol{\epsilon} = E\mathrm{tr}(A\boldsymbol{\epsilon}\boldsymbol{\epsilon}') = \sigma^2 \mathrm{tr}A$.

Step 3. So $E\hat{\sigma}^2 = \sigma^2 + \|(I - W)\mathbf{m}\|^2/\mathrm{tr}A \geq \sigma^2$. Row i of W holds the weights of the fit at X_i; for linear m and local linear W, Exercise 12.5 gives $(I - W)\mathbf{m} = 0$, so $\hat{\sigma}^2$ is unbiased. Otherwise smoothing bias inflates it. ■